Curriculum Vitae

Sheeraja Rajakrishnan

srajakri@gmail.com   sheeraja-rajakrishnan
https://sheeraja.github.io/

Research Statement

My research focuses on uncertainty-aware reinforcement learning (RL). The predictions made by machine learning models are often unreliable and tend to be overconfident. Machine learning models usually process unseen or out-of-distribution data with high confidence. In critical applications such as medical diagnoses, autonomous driving, and recommendation systems, unreliable predictions can have severe repercussions, including death, incorrect medical diagnoses, or inadequate treatment for a medical condition. If the machine learning model provided an uncertainty estimate, it would make it easier for a human to intervene and take the appropriate action to avoid these ramifications. Uncertainty estimation can guide an RL agent in learning more efficiently and much faster. My research aims to design an uncertainty-aware RL world model that is trustworthy and can achieve better performance compared to existing state-of-the-art RL world models.

Research Experience

- 2022 – Present ◇ Uncertainty-aware reinforcement learning
- 2019 ◇ Robustness of centrality measures to missing or incorrect graph data
- 2018 ◇ Collection and analysis of gait data to identify a person
- 2014 ◇ Image segmentation of heart scans to detect stenosis

Invited Talks & Guest Lectures

- September 2025 ◇ **An Introduction to Machine Learning** – Guest lecture for `SWEN101`: Software Engineering Freshman Seminar, Golisano College of Computing & Information Sciences, Rochester Institute of Technology
- February 2025 ◇ **Uncertainty in Machine Learning** – Guest lecture for `CSCI736`: Neural Networks & Machine Learning, Golisano College of Computing & Information Sciences, Rochester Institute of Technology
- July 2024 ◇ **An Introduction to Machine Learning** – Invited talk at GenCyber Summer Camp for 8th-12th grade students, Rochester Institute of Technology
- October 2023 ◇ **An Introduction to Machine Learning** – Guest lecture for `SWEN101`: Software Engineering Freshman Seminar, Golisano College of Computing & Information Sciences, Rochester Institute of Technology

Employment History

2022 – Present ◇ **PhD Candidate** Rochester Institute of Technology

- Researching on uncertainty-aware reinforcement learning models
- Passed the Research Potential Assessment in May 2022
- Successfully defended dissertation proposal in October 2025

Employment History (continued)

2020 – 2022 ◇ **Data Analyst/Data Scientist** Kaiser Permanente (DiSys)

- Analyzed video visits data and developed multiple Tableau reports to communicate the insights with the stakeholders
- Developed an interactive Tableau dashboard and custom Google Analytics reports and dashboards for analyzing COVID-19 web traffic
- Worked on a PoC algorithm for face recognition, using Python and the sklearn package
- Created a dynamic Tableau dashboard to track all the published reports on the Tableau server, using the PostgreSQL Tableau Server Repository
- Created a Tableau dashboard to forecast pageviews, using SARIMAX and the TabPy server
- Conducted data science sessions on Logistic Regression, K-Means and K-Modes Clustering, Dimensionality Reduction, Decision Trees and TabPy
- Utilized NLP models to categorize patient emails (using Python and spaCy)

2019 – 2020 ◇ **Technical Business Analyst** Kaiser Permanente (DiSys)

- Automated a portion of the testing process to reduce the testing time from 20 minutes to 30 seconds, using Python
- Worked on QA and database regression testing

2018 – 2019 ◇ **Data Analyst** Syracuse University

- Worked on the analysis and reporting of faculty research expenditures to assess the overall spending incurred by the department
- Automated all the data entry and analyses processes, to populate the data required for the analysis and reporting, using Python
- Worked on converting a manual form to an online form using JavaScript, PHP, and MySQL database

2017 – 2019 ◇ **Graduate Research Assistant** Syracuse University

- Worked on analysis of a social network using K-core decomposition, in Python, to segregate nodes into different clusters and to assess the impact of the removal of one node on other nodes

2015 – 2017 ◇ **Business Analyst** Tata Consultancy Services (TCS)

- Worked as a Lead for a few projects, in the Visual Analytics team to report the performance of the business, using Tableau
- Performed K-means clustering for customer segmentation, correlation analysis for customer spending patterns, and trend analysis on customer spending data using R, Hive, and Impala
- Created a few custom views, such as a chord chart, an arc chart, and a hexagonal map, in Tableau
- Developed an algorithm, using Impala, to validate transactions that were flagged as fraudulent transactions
- Worked, as a Lead and developer, on the full stack development of an agile project to create a web page and monitor the traffic — Created a web page, back-end databases, and reports using Impala and Tableau
- Provided training on R, Tableau, and cloud platforms to colleagues

2014 – 2015 ◇ **Trainee Decision Scientist** Mu Sigma Business Solutions Pvt. Ltd.

- Worked for the in-house Products Team to build a Naive Bayes Classifier Package, using R, to perform classification of textual and non-textual data
- Worked with Business Intelligence team, of a major multinational mass media and information client, to perform data analyses and generate analytic reports using SAP Business Objects and Tableau

Education

2022 – 05/27 (Expected) ◇ **Ph.D., Rochester Institute of Technology** Computing and Information Sciences

2017 – 2019 ◇ **M.S., Syracuse University** Computer Science

2010 – 2014 ◇ **B.E., SSN College of Engineering** Computer Science

Skills

Languages	◇ Python (PyTorch, TensorFlow), C++, C
Front End	◇ HTML, CSS, PHP
Software	◇ Tableau, Jupyter Notebook, Google Analytics, RStudio, Eclipse, SAP BO
Databases	◇ MySQL, Hive, Impala
Version Control	◇ Git
Familiar with	◇ Python (Jax), Julia, R, SAS, Haskell, Weka, Stata, MATLAB
Misc.	◇ Academic research, teaching/training, LaTeX typesetting